

Teacher Information, Course Description and
Lecture-wise Plan

of

Data Structures (CSC211)

under

Pilot Project:

Uniform Teaching, Assessment and Quality
Assurance Across All the Campuses

by

Faculty of Information Science and Technology
COMSATS University Islamabad

A: Teacher Information

Course Name: CSC211 Discrete Structures	Credit Hours: 3+1
Pre-requisite:	CSC103-Programming Fundamentals
Teacher Name:	Teacher E-mail:
Program: BS (Computer Science)	Semester: Fall 2026
Department: Computer Science	Campus:

B: Course Contents

This course provides fundamental knowledge of data organization. The topics include Foundations of Data Structures; Linear Data Structures; Non-Linear Data Structures; Hashing & Efficient Data Access; and Searching, Sorting & Optimization.
Recommended Books: 1. Data Structures & Algorithms in C++ , Goodrich, Tamassia, Mount & Goldwasser, 2024 2. Data Structures and Algorithm Analysis in C++, Mark Allen Weiss, 2024.

C: Lecture-wise Plan

Week #	Lecture #	Topic (Chapter/Topics Covered)	Practice/ Evaluation	Remarks
1	1 (Date)	Introduction to Data Structures: • Course Overview • Learning Outcomes and Course Expectations • Data and Information • Data Types • Data Structures • Algorithms • Relationship among Data, Data Structures and Algorithms • Need for Data Structures • Applications of Data Structures	Practice/ Identify suitable data structures for given scenarios Evaluation Short conceptual and scenario-based questions	Goodrich Ch. 1 Weiss Ch. 1
	2 (Date)	Classification & Operations of Data Structures: • Criteria for Classification of Data Structures • Primitive and Non-Primitive Data Structures • Linear and Non-Linear Data Structures • Static and Dynamic Data Structures • Sequential and Random Access • Persistent and Ephemeral Data Structures • Operations of Data Structures: (Traversal, Insertion, Deletion, Searching, Sorting, Updating, Merging) Abstract Data Types & Data Structure Design • Abstract Data Type (ADT) • Interface and Implementation • Abstraction • Encapsulation • Trade-offs in Data Structure Design • Time-Space Trade-off • Simplicity versus Efficiency • Criteria for Selecting Appropriate Data Structures • Impact of Evolving Application Requirements	Classify given data structures and identify appropriate operations	Goodrich Ch. 3 Weiss: Ch. 3
2	3 (Date)	Algorithm Analysis & Complexity • Algorithm Efficiency • Basic Operations • Time Complexity • Space Complexity • Best-Case Analysis • Average-Case Analysis • Worst-Case Analysis • Asymptotic Analysis • Big-O Notation • Big-Theta Notation	Quiz # 1	Goodrich Ch. 4 Weiss: Ch. 4

		• Big-Omega Notation • Common Complexity Classes		
	4 (Date)	Arrays & Sequential Storage: • Array Concept • Memory Representation • Indexing • One-Dimensional Arrays • Two-Dimensional Arrays • Operations on Array: (Traversal, Insertion, Deletion, Searching, Updating) • Advantages and Limitations of Arrays • Complexity of Array Operations	Perform array operations and trace 1D/2D array indexing	Goodrich Ch. 2–3 Weiss: Ch. 3
3	5 (Date)	Linked Lists: Fundamentals: • Need for Linked Lists • Linked List Concept • Memory Organization • Linked List ADT • Node Structure • Head and Tail • Creation of Linked List • Traversal of Linked List • Basic Insertion • Basic Deletion • Advantages and Limitations of Linked Lists	Create, traverse, insert, and delete nodes in a singly linked list	Goodrich Ch. 3 Weiss: Ch. 3
	6 (Date)	Singly Linked List Operations: • Singly Linked List • Linked List Operations (Insertion at Beginning, Insertion at End, Insertion at Specific Position, Deletion from Beginning, Deletion from End, Deletion from Specific Position, Searching in Singly Linked List, Updating a Node, Counting Nodes, Reversing a Linked List) • Complexity of Singly Linked List Operations • Practice and Evaluation	+ Assignment No 1	Goodrich Ch. 3 Weiss: Ch. 3
4	7	Doubly Linked Lists: Concepts and Operations: • Doubly Linked List Concept • Node Structure: Previous, Data and Next • Operations: (Forward Traversal, Backward Traversal, Insertion at the Beginning, Insertion at the End, Insertion at a Specific Position, Deletion from the Beginning, Deletion from the End, Deletion from a Specific Position, Searching and Updating) • Applications of Doubly Linked Lists	Quiz # 2	Goodrich Ch. 3 Weiss: Ch. 3

		• Practice and Evaluation		
	8	Circular Linked List Concept • Circular Node Structure • Head and Tail in Circular Linked Lists • Operations on Circular List (Traversal, Insertion at the Beginning, Insertion at the End, Insertion at a Specific Position, Deletion from the Beginning, Deletion from the End, Deletion from a Specific Position) • Applications of Circular Linked Lists • Comparison of Singly, Doubly and Circular Linked Lists • Practice and Evaluation	Perform traversal, insertion, and deletion in circular linked lists	
5	9 (Date)	Linear Structures: Applications & Problem Solving: • Applications of Linked Lists • Dynamic Data Management • Traversal and Manipulation • Selection of Appropriate Linear Data Structures • Array-Based Problem Solving • Linked-List-Based Problem Solving • Combined Linear Structure Operations • Real-World Applications	Solve a problem requiring selection between an array and linked list for dynamic data management	Goodrich Ch. 3 Weiss: Ch. 3
	10 (Date)	Stack ADT & Implementation: • Stack Concept • LIFO Principle • Stack ADT • Operation on Stack (Push, Pop, Peek/Top) • Array-Based Implementation of Stack • Linked-List Implementation of Stack • Complexity of Stack Operation	Implement and trace push, pop, and peek operations using array-based and linked-list stacks	Goodrich Ch. 6 Weiss: Ch. 3
6	11 (Date)	Stack Applications: Expression Conversion: • Applications of Stacks - Function Call Management - Backtracking - Undo and Redo Operations Expression Notation • Infix Notation • Prefix Notation • Postfix Notation • Operator Precedence • Operator Associativity • Infix-to-Postfix Conversion	Convert infix expressions to postfix and evaluate postfix expressions using a stack	Goodrich Ch. 6 Weiss: Ch. 3

		• Postfix Expression Evaluation		
	12 (Date)	Recursion Concept • Basic Concept • Base Case • Recursive Case • Call Stack • Recursion versus Iteration • Types of Recursion • Recursive Problem Solving • Advantages and Limitations of Recursion	Quiz # 3	Goodrich Ch. 5–6 Weiss: Ch. 1, 3
7	13 (Date)	Queue ADT & Implementation: • Queue Concept • FIFO Principle • Queue ADT • Enqueue Operation • Dequeue Operation • Array-Based Implementation • Linked-List Implementation • Complexity of Queue Operations	Implement and trace enqueue and dequeue operations using array-based and linked-list queues	Goodrich Ch. 6 Weiss: Ch. 3
	14 (Date)	Circular Queue, Deque & Priority Queue: • Limitations of Simple Queue Circular Queue Concept • Advantages of Circular Queue Deque Concept • Insertion at Both Ends • Deletion at Both Ends • Input-Restricted Deque • Output-Restricted Deque • Priority Queue Concept • Priority-Based Processing	Trace circular queue, deque, and priority queue operations for given sequences	Goodrich Ch. 6, 9 Weiss: Ch. 3, 6
8	15 (Date)	Queue Applications & Linear Structure Selection: • Job Scheduling • Printer Spoolers • Router and Network Buffers • Resource Management • Simulation Applications • Selecting Appropriate Linear Data Structures • Problem Solving Using Linear Structures	+ Assignment No 2	Goodrich Ch. 6, 9 Weiss: Ch. 3
	16	Trees & General Trees:		

	(Date)	• Need for Non-Linear Data Structures • Tree Concept • Basic Tree Terminology: (Root, Node, Edge, Parent, Child, Sibling, Leaf Node, Internal Node, Degree, Depth, Height, Subtree General Trees) • Introduction to Tree Representation • Binary Tree Concept • Binary Tree Properties • Types of Binary Trees: Full, Complete and Skewed • Sequential/Array Representation of Binary Trees • Linked Representation of Binary Trees	Identify tree components and construct binary trees using array and linked representations	Goodrich Ch. 8 Weiss: Ch. 4
9	17	Binary Trees & Representation • Binary Tree ADT • Binary Tree Concept • Binary Tree Properties • Full Binary Tree • Complete Binary Tree • Skewed Binary Tree • Sequential/Array Representation • Linked Representation • Binary Tree Implementation	Construct binary trees using array and linked representations	Goodrich Ch. 8 Weiss: Ch. 4
	18 (Date)	Midterm Examination	Midterm (25%)	
10 10	19 (Date)	Binary Tree Traversals: • Tree Traversal Concept • Pre-Order Traversal • In-Order Traversal • Post-Order Traversal • Recursive Traversal • Level-Order Traversal • Comparison of Traversal Methods • Applications of Tree Traversal	Perform and trace Pre-order, In-order, Post-order, and Level-order traversals	Goodrich Ch. 8 Weiss: Ch. 4
	20 (Date)	Binary Search Trees: Search & Insertion: • Binary Search Tree Concept • BST Property • Binary Tree versus BST • Searching in BST • Recursive Search	Construct a BST and perform iterative and recursive search operations	Goodrich Ch. 11 Weiss: Ch. 4

		• Iterative Search • Insertion in BST • Minimum Value • Maximum Value • Complexity of BST Operations • Applications of BST		
11	21 (Date)	Binary Search Trees: Deletion & Analysis • BST Deletion • Deletion of Leaf Node • Deletion of Node with One Child • Deletion of Node with Two Children • Balanced BST • Skewed BST • Best-Case and Worst-Case Behavior • Applications of BST	Perform BST deletion for leaf, one-child, and two-child cases	Goodrich Ch. 11 Weiss: Ch. 4
	22 (Date)	AVL Trees: • Need for Balanced Search Trees • AVL Tree Concept • Balance Factor • Height Balance • AVL Tree Rotations (Single Rotations, Left Rotation, Right Rotation, Double Rotations) • Operations on AVL Tree (Insertion, Searching, Deletion) • AVL Tree Performance	+ Assignment No 3	Goodrich Ch. 11 Weiss: Ch. 4
12	23 (Date)	Heap & Priority Queue: • Heap Concept • Heap Property • Types of Heap (Min-Heap, Max-Heap) • Complete Binary Tree Representation • Array Representation of Heap • Operations on Heap (Insertion, Deletion ,Heapify, Build-Heap) • Heap-Based Priority Queue • Heap Sort	Construct a min/max heap and perform insertion, deletion, heapify, and build-heap operations	Goodrich Ch. 9 Weiss: Ch. 6
	24 (Date)	Tree Applications: • Decision Trees • Expression Trees • Huffman Tree • Huffman Coding • Applications of Binary Trees	Quiz # 4	Goodrich Ch. 9, 13 Weiss: Ch. 4, 7

13	25 (Date)	Graphs & Graph Representation: • Graph Concepts: (Vertices, Edges, Directed, Graphs, Undirected Graphs, Weighted Graphs Unweighted Graphs, Degree, Path, Cycle, Connected Graph) • Types of Graphs • Adjacency Matrix • Adjacency List • Comparison of Graph Representations	Construct graphs and represent them using adjacency matrix and adjacency list	Goodrich Ch. 14 Weiss: Ch. 9
	26 (Date)	Graph Traversal: BFS & DFS: • Graph Traversal Concept • Breadth-First Search (BFS) • BFS Algorithm • Depth-First Search (DFS) • DFS Algorithm • BFS versus DFS • Traversal Complexity • Applications of BFS and DFS	Perform BFS and DFS traversals on given graphs and trace the visited order	Goodrich Ch. 14 Weiss: Ch. 9
14	27 (Date)	Shortest Path: Dijkstra's Algorithm • Shortest-Path Concept • Single-Source Shortest Path • Dijkstra's Algorithm • Time Complexity • Applications • Limitations of Dijkstra's Algorithm	Apply Dijkstra's algorithm to find shortest paths in a weighted graph	Goodrich Ch. 14 Weiss: Ch. 9
	28 (Date)	Minimum Spanning Trees • Spanning Tree Concept • Minimum Spanning Tree • Prim's Algorithm • Kruskal's Algorithm • Comparison of Prim and Kruskal • Applications of MST	Construct an MST using Prim's and Kruskal's algorithms	Goodrich Ch. 14 Weiss: Ch. 9
15	29 (Date)	Searching Algorithms • Searching Problem • Linear Search • Linear Search Complexity • Binary Search (Iterative, Recursive) • Binary Search Complexity • Comparison of Linear and Binary Search	+ Assignment No 4	Goodrich Ch. 4, 13 Weiss: Ch. 2

		• Impact of Data Organization on Searching		
	30 (Date)	Sorting: Elementary Algorithms: • Sorting Problem • Characteristics of Sorting Algorithms • Bubble Sort • Selection Sort • Insertion Sort • In-Place Sorting • Comparison of Elementary Sorting Algorithms	Trace Bubble, Selection, and Insertion Sort on given arrays	Goodrich Ch. 13 Weiss: Ch. 7
16 16	31 (Date)	Sorting: Efficient Algorithms: • Divide-and-Conquer Strategy • Merge Sort • Quick Sort • Comparison of Merge Sort and Quick Sort	Trace Merge Sort and Quick Sort on given arrays using divide-and-conquer	Goodrich Ch. 13 Weiss: Ch. 7
	32 (Date)	Hashing & Efficient Data Access • Need for Efficient Data Access • Hashing Concept • Hash Table • Key-Value Mapping • Hash Function • Properties of Good Hash Functions • Open and Closed Hashing • Collision • Separate Chaining • Linear Probing • Quadratic Probing • Double Hashing • Rehashing • Applications of Hashing	Construct a hash table and resolve collisions using chaining and probing	Goodrich Ch. 10 Weiss: Ch. 5
17	(Date)		Final (50%)	

Note: All Assignments/Quizzes weightage is 25%